

XYLOFY AI

Week 1 Internship Project Assignment from XYlofy AI

MACHINE LEARNING PROJECT REPORT

House Price Prediction

Name : **Ravi kumar Yadav**

College : **Baba Farid College of Engineering & Technology**

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1. Executive Summary

This project investigates the prediction of residential house prices using structured property attributes such as area, number of bedrooms and bathrooms, number of stories, parking capacity, and a range of qualitative amenities (main-road access, air conditioning, basement, hot-water heating, guest room, and preferred area status). The objective is to build, evaluate, and compare two supervised machine learning regression models — Linear Regression and Random Forest Regression — to determine which approach more accurately predicts house prices on this dataset, and which property features most strongly influence price.

The analysis was carried out in Python within a Jupyter Notebook environment, using the Housing.csv dataset comprising 545 property records across 13 attributes. Categorical variables were label-encoded, the data was split into training (80%) and testing (20%) subsets, and both models were trained on identical data partitions to ensure a fair comparison.

The results show that the Linear Regression model achieved a higher R^2 score (0.592) than the Random Forest Regression model (0.546) on the held-out test set, indicating that, for this dataset and feature set, the simpler linear model generalised slightly better than the ensemble model. Both models identified built-up area as the single most influential factor in determining price, followed by the number of bathrooms and the presence of amenities such as air conditioning. This report documents the complete workflow — from data understanding and preprocessing through model building, evaluation, and visual analysis — and concludes with key insights and recommendations for further improvement.

1.1 Key Findings at a Glance

- Linear Regression outperformed Random Forest Regression on this dataset, achieving an R^2 score of 0.592 versus 0.546.
- Property area is the dominant predictor of price in both models, followed by bathrooms, air conditioning, and stories.
- The dataset contains no missing values, simplifying the preprocessing pipeline.
- Average prediction error (MAE) is approximately ₹8.4 lakh (Linear Regression) and ₹8.6 lakh (Random Forest) against a mean house price of roughly ₹47.7 lakh.
- House prices in the dataset are right-skewed, with most properties priced between ₹25 lakh and ₹60 lakh, and a long tail of higher-value properties.

2. Introduction

2.1 Background

Accurately predicting the market price of a house is a problem of significant practical value to buyers, sellers, real-estate agents, and financial institutions. Traditional valuation relies heavily on manual comparison with similar properties and expert judgement, which can be inconsistent and slow. Machine learning offers a data-driven alternative: by learning the statistical relationship between measurable property characteristics and historical sale prices, a trained model can produce fast, repeatable, and objective price estimates for new listings.

This project applies two widely used regression techniques — Linear Regression, a classical statistical method, and Random Forest Regression, an ensemble machine learning method based on decision trees — to a housing dataset, with the aim of comparing their predictive accuracy and interpretability.

2.2 Objectives

1. To explore and understand the structure, distribution, and relationships within the housing dataset.
2. To preprocess the data, including encoding categorical variables, so that it is suitable for machine learning algorithms.
3. To build and train a Linear Regression model and a Random Forest Regression model on the same training data.
4. To evaluate and compare both models using standard regression metrics: R^2 score, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE).
5. To identify which property features have the greatest influence on predicted house price.
6. To visually communicate the dataset characteristics and model performance through charts and graphs.

2.3 Scope

The scope of this project is limited to supervised regression on a single, static dataset of 545 housing records. The project does not incorporate external data sources (such as location coordinates, school quality, or macroeconomic indicators), nor does it explore advanced techniques such as hyperparameter tuning, cross-validation, regularisation (Ridge/Lasso), or gradient-boosted models. These are identified as opportunities for future work in the concluding section of this report.

2.4 Tools and Technologies Used

Category	Tools / Libraries
Programming Language	Python 3

Category	Tools / Libraries
Development Environment	Jupyter Notebook
Data Handling	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Machine Learning	Scikit-learn (LinearRegression, RandomForestRegressor, LabelEncoder, train_test_split, metrics)

3. Dataset Description

The dataset used for this project, `Housing.csv`, contains 545 records describing residential properties along with their sale prices. Each row represents a single property, and each column captures a distinct attribute of that property. The dataset combines numerical, binary categorical, and multi-class categorical variables, making it a representative example of real-world tabular data requiring preprocessing before model training.

3.1 Dataset Overview

Total Records	545
Total Attributes	13 (1 target + 12 predictors)
Missing Values	None — dataset is complete
Target Variable	price (continuous, in currency units)
Numeric Predictors	area, bedrooms, bathrooms, stories, parking
Categorical Predictors	mainroad, guestroom, basement, hotwaterheating, airconditioning, prefarea, furnishingstatus

3.2 Attribute Dictionary

Attribute	Type	Description
price	Numeric (target)	Sale price of the property
area	Numeric	Total built-up area of the property (square feet)
bedrooms	Numeric	Number of bedrooms
bathrooms	Numeric	Number of bathrooms
stories	Numeric	Number of stories/floors in the house
mainroad	Categorical (yes/no)	Whether the house is connected to the main road
guestroom	Categorical (yes/no)	Whether the house has a guest room
basement	Categorical (yes/no)	Whether the house has a basement
hotwaterheating	Categorical (yes/no)	Whether the house has hot water heating
airconditioning	Categorical (yes/no)	Whether the house has air conditioning
parking	Numeric	Number of parking spaces

Attribute	Type	Description
prefarea	Categorical (yes/no)	Whether the house is in a preferred/desirable area
furnishingstatus	Categorical (3-class)	Furnished, semi-furnished, or unfurnished

3.3 Summary Statistics (Numeric Features)

Feature	Mean	Std Dev	Min	Max
price	4,766,729	1,870,440	1,750,000	13,300,000
area	5,150.5	2,170.1	1,650	16,200
bedrooms	2.97	0.74	1	6
bathrooms	1.29	0.50	1	4
stories	1.81	0.87	1	4
parking	0.69	0.86	0	3

4. Methodology

The project followed a standard supervised machine learning workflow: data loading, exploratory analysis, preprocessing, train-test splitting, model training, prediction, and evaluation. Each stage is described below, with reference to the corresponding operations performed in the Jupyter Notebook.

4.1 Data Loading and Inspection

The dataset was loaded into a Pandas DataFrame using `pd.read_csv('Housing.csv')`. Initial inspection used `df.shape` to confirm dimensions (545 rows × 13 columns), `df.info()` to review data types, and `df.isnull().sum()` to confirm there were no missing values across any column. This clean starting point meant no imputation strategy was required.

4.2 Feature and Target Separation

The target variable, price, occupies the first column of the dataset and was separated as `y = df.iloc[:, 0]`. The remaining 12 columns were assigned to `x = df.iloc[:, 1:]` as candidate predictor variables.

4.3 Encoding Categorical Variables

The predictor set contains a mix of numeric and text-based categorical columns. The five purely numeric columns (area, bedrooms, bathrooms, stories, parking) were isolated directly. The six categorical text columns (mainroad, guestroom, basement, hotwaterheating, airconditioning, prefarea) were transformed using Scikit-learn's `LabelEncoder`, which converts each category (for example, "yes"/"no") into a corresponding integer code (1/0). The encoded categorical columns were then concatenated with the numeric columns using `pd.concat()` to form the final feature matrix `X`, containing 11 columns in total.

Note: in the version of the notebook reviewed, the `furnishingstatus` column (a three-class categorical variable) was not included among the encoded columns used to build the final feature matrix; the model was therefore trained on 11 of the 12 available predictor columns. This is discussed further in the Limitations section of this report.

4.4 Train-Test Split

The feature matrix `X` and target vector `y` were split into training and testing subsets using `train_test_split()` with a `test_size` of 0.2 (20%) and a fixed `random_state` of 41 for reproducibility. This resulted in 436 records for training and 109 records for testing, ensuring the models were evaluated on data they had not seen during training.

Subset	Number of Records	Proportion
Training Set	436	80%
Testing Set	109	20%

4.5 Model Selection

Two regression algorithms were selected to provide a comparison between a simple, interpretable linear method and a more flexible, non-linear ensemble method:

- **Linear Regression:** A statistical model that assumes a linear relationship between the predictor variables and the target. It is fast to train, easy to interpret through its coefficients, but limited in its ability to capture non-linear relationships.
- **Random Forest Regression:** An ensemble learning method that builds multiple decision trees on random subsets of data and features, averaging their predictions. It can capture non-linear relationships and interactions between features, at some cost to interpretability. The model was configured with 200 estimators (`n_estimators=200`) and a fixed `random_state` of 42.

4.6 Evaluation Metrics

Three standard regression metrics were used to evaluate and compare model performance on the test set:

- **R² Score (Coefficient of Determination):** Measures the proportion of variance in the target variable explained by the model. Ranges from 0 to 1 (higher is better), where 1 indicates perfect prediction.
- **Mean Absolute Error (MAE):** The average absolute difference between predicted and actual prices, in the same currency units as the target variable. Lower is better.
- **Root Mean Squared Error (RMSE):** Similar to MAE but squares errors before averaging and then takes the square root, which penalises larger errors more heavily. Lower is better.

5. Exploratory Data Analysis

Before building predictive models, the dataset was explored visually to understand the distribution of house prices and the relationships between numeric features. This step helps validate assumptions, detect skewness or outliers, and identify which features are likely to be useful predictors.

5.1 Distribution of House Prices

The histogram below shows the distribution of the price variable across all 545 properties. The distribution is right-skewed: the majority of properties are priced between approximately ₹2.5 million and ₹6 million, with a peak frequency around ₹3.5–4 million. A smaller number of higher-value properties extend the tail of the distribution out to approximately ₹13.3 million. This skewness is typical of real-world price data and suggests that a small number of premium properties pull the mean price (₹4.77 million) above the median.

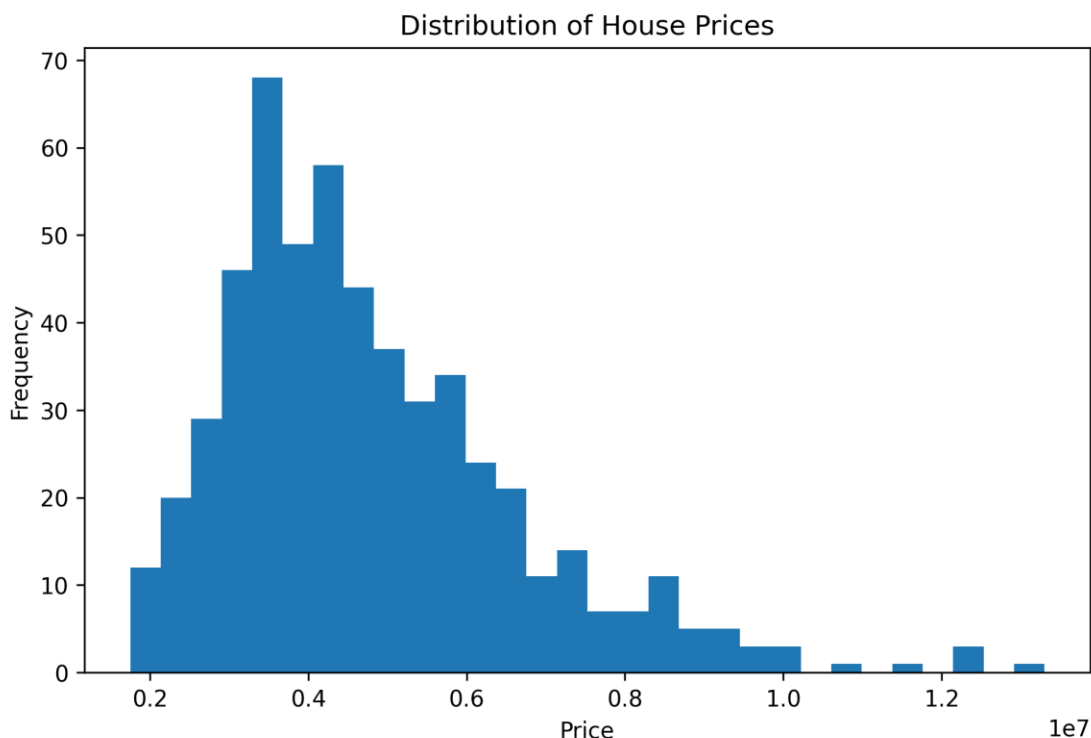


Figure 5.1: Distribution of House Prices across the dataset

5.2 Correlation Analysis

A correlation heatmap was generated to examine the pairwise linear relationships between all numeric and encoded categorical features. Correlation coefficients range from -1 (strong negative relationship) to +1 (strong positive relationship), with values near 0 indicating little to no linear relationship.

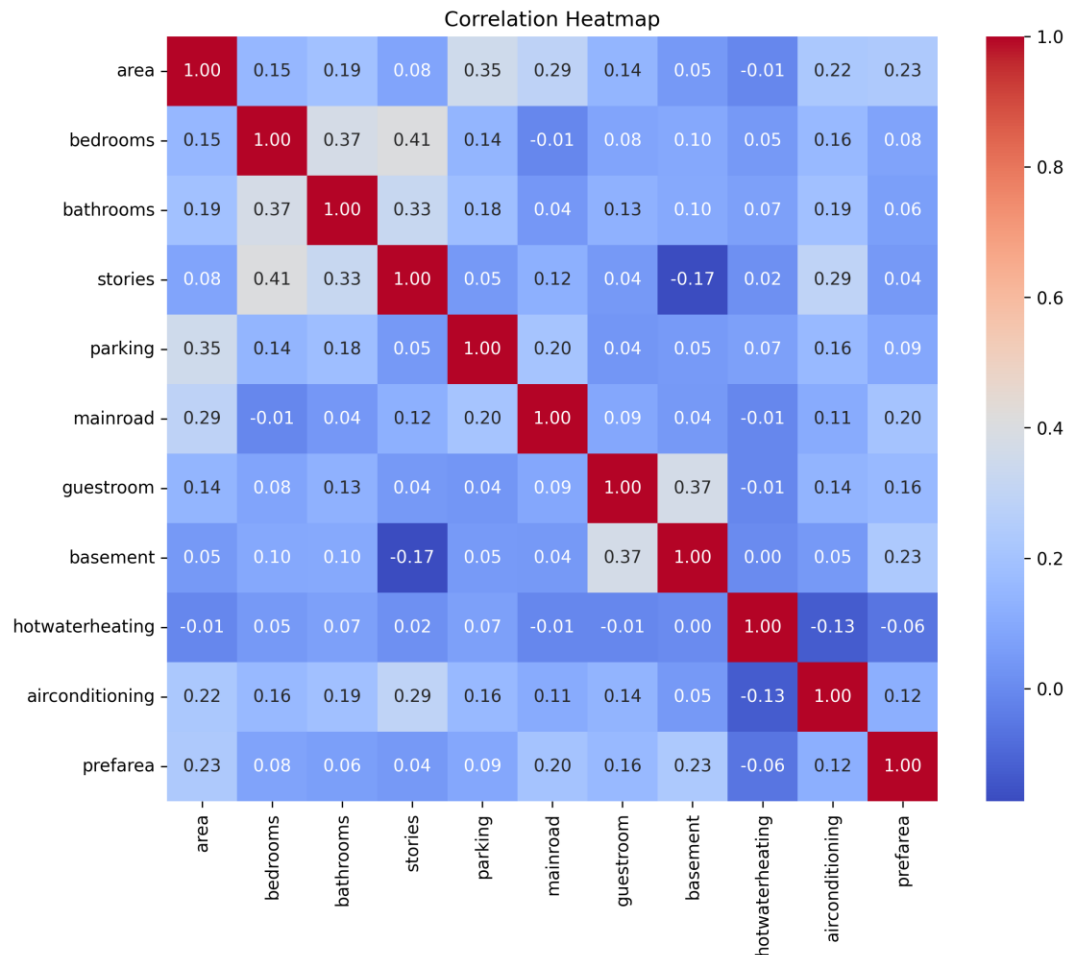


Figure 5.2: Correlation Heatmap of all features used in modelling

Several observations emerge from the heatmap. Bedrooms and stories show a moderate positive correlation (0.41), as do bedrooms and bathrooms (0.37) and guestroom and basement (0.37), suggesting these amenities tend to co-occur in larger properties. Area shows a mild positive correlation with parking (0.35) and main-road access (0.29). No pair of features shows extremely high correlation (above 0.5), which suggests multicollinearity is not a severe concern for the Linear Regression model in this dataset.

6. Model Building and Results

6.1 Linear Regression Model

The Linear Regression model was trained on the 436-record training set using Scikit-learn's LinearRegression class with default parameters. After training, the model was used to predict prices for the 109-record test set, and predictions were compared against actual values using the three evaluation metrics described in Section 4.6.

Metric	Value
R ² Score	0.5919
Mean Absolute Error (MAE)	₹842,170.60
Root Mean Squared Error (RMSE)	₹1,140,983.09

An R² score of 0.5919 indicates that the Linear Regression model explains approximately 59% of the variance in house prices using the available features. The remaining variance is likely attributable to factors not captured in the dataset, such as exact location, neighbourhood quality, property age and condition, and prevailing market conditions at time of sale.

6.1.1 Feature Importance (Linear Regression Coefficients)

The bar chart below shows the coefficient assigned to each feature by the Linear Regression model. A coefficient represents the estimated change in price (in currency units) associated with a one-unit increase in that feature, holding all other features constant. Larger positive coefficients indicate a stronger positive association with price.

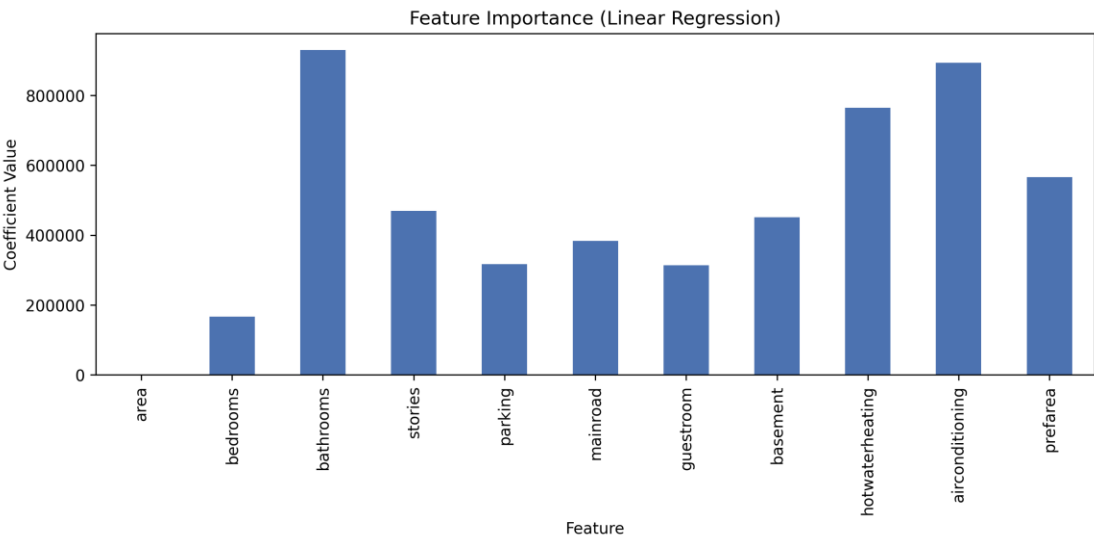


Figure 6.1: Feature coefficients from the Linear Regression model

Air conditioning (₹893,605), hot water heating (₹764,332), and bathrooms (₹930,700) carry the largest positive coefficients, indicating that homes with these features command substantially higher prices. Area has a small per-unit coefficient (₹273.75) because it is measured in square feet, where each individual square foot contributes a modest amount, but the cumulative effect across thousands of square feet is large.

6.2 Random Forest Regression Model

The Random Forest Regression model was trained on the same training set using 200 decision trees (`n_estimators=200`) with a fixed random seed (`random_state=42`) for reproducibility. Random Forest builds many decision trees on bootstrapped samples of the data and averages their individual predictions, which typically reduces overfitting compared to a single decision tree.

Metric	Value
R ² Score	0.5463
Mean Absolute Error (MAE)	₹855,481.94
Root Mean Squared Error (RMSE)	₹1,203,100.96

The Random Forest model achieved an R² score of 0.5463, slightly lower than the Linear Regression model. This is a somewhat counter-intuitive result, since ensemble tree-based models often outperform linear models on tabular data with non-linear relationships. Possible explanations include the relatively small size of the training set (436 records), the use of default/unt-used hyperparameters, and the dataset's predominantly linear or monotonic relationships between features and price, which would favour a linear model.

6.2.1 Feature Importance (Random Forest)

Random Forest provides a feature importance score for each predictor, based on how much each feature reduces prediction error (impurity) across all trees in the forest. Unlike linear coefficients, these importance scores are always non-negative and sum to 1 across all features, making them directly comparable.

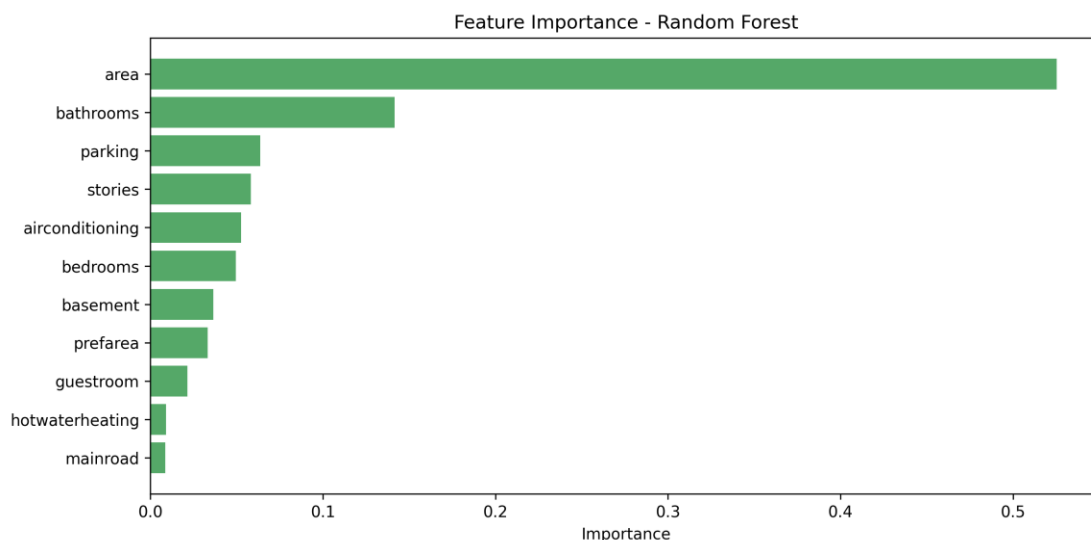


Figure 6.2: Feature importance ranking from the Random Forest model

Area dominates the Random Forest's decision-making, accounting for approximately 52.5% of total feature importance, more than the next six features combined. Bathrooms (14.2%) and parking (6.4%) follow as the next most influential features. This aligns with the Linear Regression coefficients in confirming that area and bathrooms are consistently important, though the relative ranking of other features differs between the two models.

7. Model Comparison

To directly compare the two models, their evaluation metrics are presented side by side below, along with the visualizations generated for this purpose.

Metric	Linear Regression	Random Forest
R ² Score	0.592 (Higher — Better)	0.546
MAE (₹)	842,171 (Lower — Better)	855,482
RMSE (₹)	1,140,983 (Lower — Better)	1,203,101

7.1 Model Accuracy Comparison

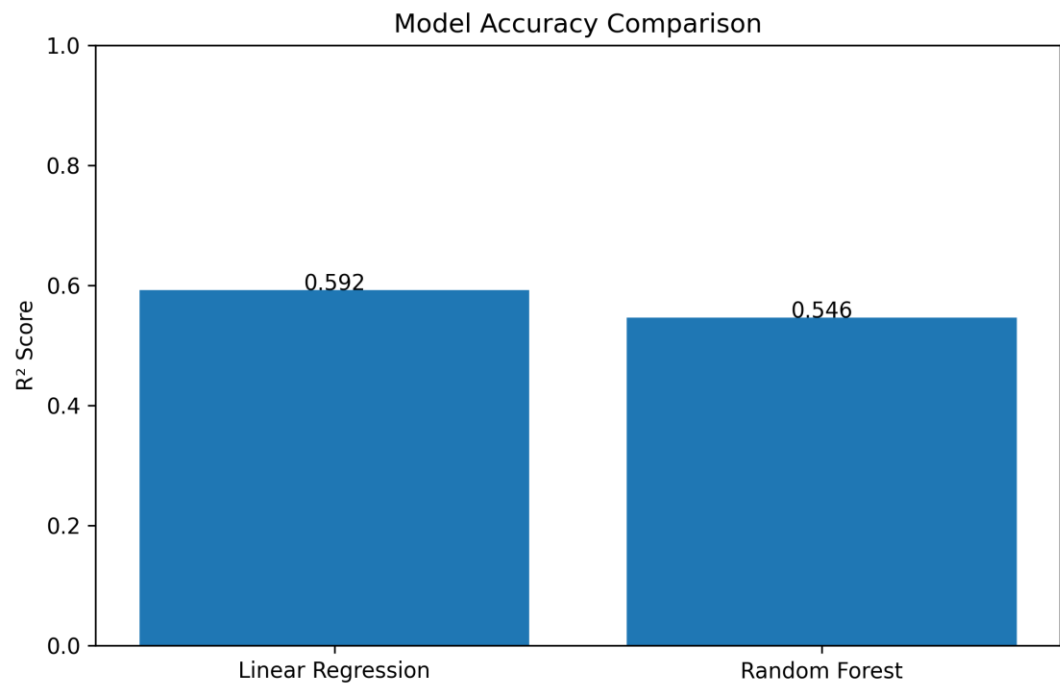


Figure 7.1: R² score comparison between Linear Regression and Random Forest

7.2 Combined Metrics Comparison

The chart below presents all three evaluation metrics — R² score, MAE, and RMSE — for both models side by side, providing a complete view of relative performance.

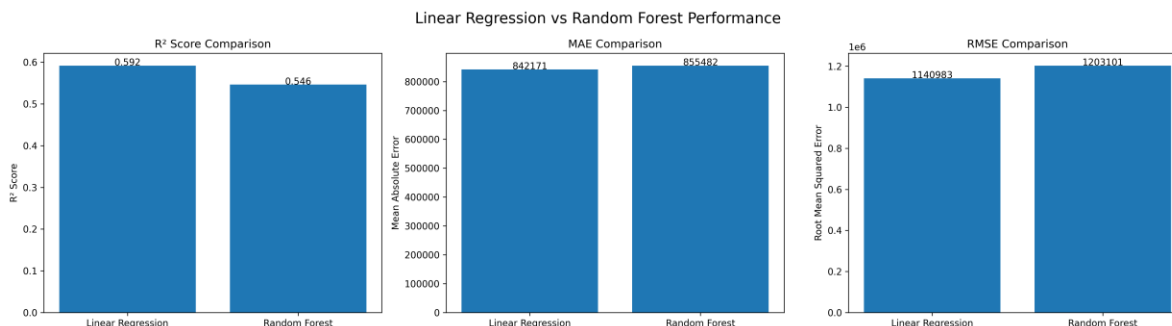


Figure 7.2: Combined comparison of R^2 , MAE, and RMSE across both models

7.3 Actual vs. Predicted Prices

The scatter plot below compares actual test-set prices against predicted prices. Each point represents one property in the test set; the red dashed line represents perfect prediction (where predicted price equals actual price). Points lying close to this line indicate accurate predictions, while points far from the line indicate larger errors.

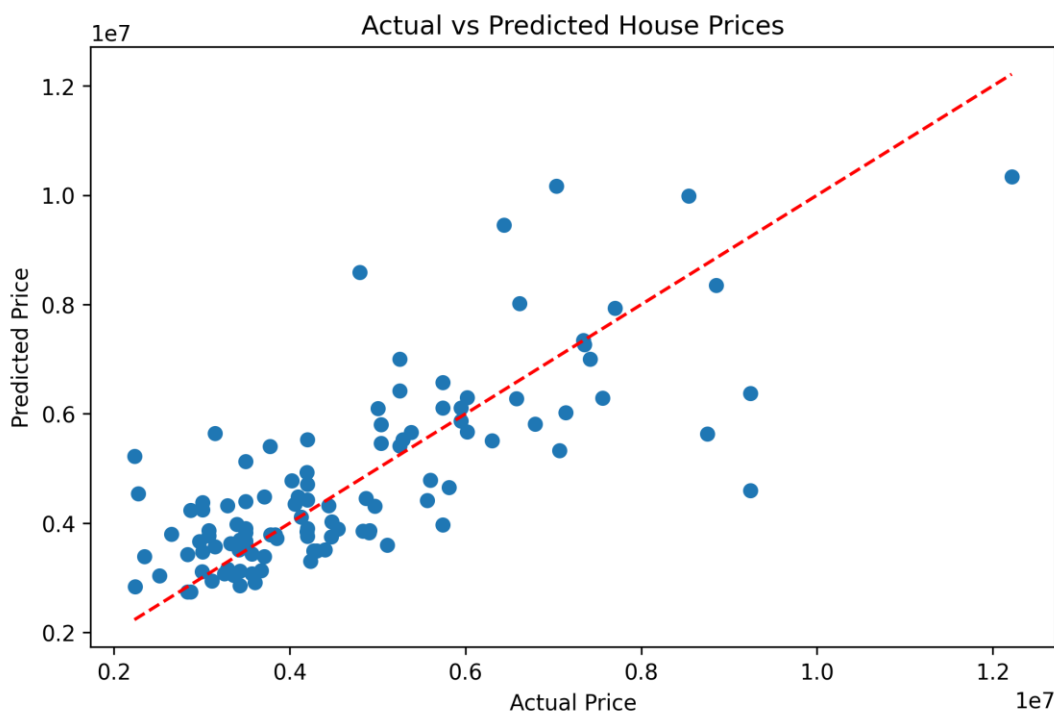


Figure 7.3: Actual vs Predicted house prices on the test set

The scatter plot shows a reasonably strong positive relationship between actual and predicted prices, with most points clustered around the line of perfect prediction at lower and mid price ranges (below ₹6 million). Prediction accuracy appears to decline somewhat for higher-priced properties, where the model tends to either over- or under-predict more substantially — a common challenge given the relatively small number of high-value properties available for the model to learn from.

7.4 Interpretation

Overall, the Linear Regression model is the better-performing model on this dataset by all three metrics, although the margin is modest (an R^2 difference of approximately 0.046). Both models broadly agree on the importance of area and bathrooms as leading predictors of price, which strengthens confidence in this finding. Neither model achieves a very high R^2 (above 0.8 or 0.9), which indicates that a substantial portion of price variation remains unexplained by the available features — a result that is typical for housing datasets lacking location-specific information.

8. Conclusion and Recommendations

8.1 Summary of Findings

- The Linear Regression model outperformed the Random Forest Regression model on this dataset, achieving an R^2 score of 0.592 compared to 0.546, along with lower MAE and RMSE.
- Built-up area is the single most important predictor of house price across both models, consistent with intuitive real-estate principles.
- Amenities such as air conditioning, hot water heating, and the number of bathrooms also exert a strong positive influence on predicted price.
- Both models leave a substantial share of price variance unexplained (40–45%), suggesting the current feature set does not capture all factors that drive real-world housing prices, such as exact location, property age, or neighbourhood-level dynamics.

8.2 Limitations

- The furnishingstatus column, a potentially informative three-category feature, was not included in the final feature matrix used for model training.
- Neither model's hyperparameters were tuned (e.g., via grid search or cross-validation); both were trained using default or arbitrarily chosen settings.
- No cross-validation was performed; results are based on a single train-test split, which may not fully represent model performance on unseen data.
- The dataset does not include location-specific variables (such as locality, city zone, or proximity to amenities), which are typically strong predictors of real-estate price.
- The relatively small dataset size (545 records) limits the ability of more complex models like Random Forest to realise their full potential.

8.3 Recommendations for Future Work

1. Include the furnishingstatus feature (properly encoded, e.g. via one-hot encoding) in the model to assess its impact on predictive accuracy.
2. Apply k-fold cross-validation to obtain a more robust and reliable estimate of model performance.
3. Perform hyperparameter tuning for the Random Forest model (e.g., tuning max_depth, min_samples_split, and n_estimators via GridSearchCV or RandomizedSearchCV) to assess whether it can be made to outperform Linear Regression.
4. Experiment with additional algorithms such as Gradient Boosting, XGBoost, or Ridge/Lasso regularised linear models for further comparison.
5. Incorporate additional real-world features, such as location/locality, property age, and proximity to amenities, if such data becomes available, to improve explanatory power.
6. Apply log-transformation to the price variable to address its right-skewed distribution, which may improve the performance of the Linear Regression model in particular.

8.4 Final Remarks

This project successfully demonstrates an end-to-end machine learning workflow for a regression problem: from raw data exploration through preprocessing, model training, evaluation, and comparison. While the Linear Regression model proved marginally more accurate than Random Forest Regression on this particular dataset, the analysis underscores that model performance is highly dependent on the size, quality, and completeness of the underlying feature set, rather than model complexity alone. The visualizations and feature importance analyses provide actionable insight into which property characteristics most strongly drive house prices, offering a useful foundation for further refinement and real-world application.

9. Appendix

9.1 Project File Inventory

File Name	Description
Housing.csv	Source dataset (545 records, 13 columns)
house_price_prediction.ipynb	Jupyter Notebook containing the full analysis and modelling code
house_price_histogram.png	Chart: distribution of house prices
correlation_heatmap.png	Chart: correlation heatmap of dataset features
actual_vs_predicted.png	Chart: actual vs predicted prices scatter plot
model_accuracy_comparison.png	Chart: R^2 score comparison bar chart
model_metrics_comparison.png	Chart: combined R^2 / MAE / RMSE comparison

9.2 Key Python Libraries Referenced

- pandas — data loading and manipulation
- numpy — numerical computation
- matplotlib.pyplot — data visualization
- seaborn — statistical visualization (correlation heatmap)
- sklearn.preprocessing.LabelEncoder — categorical variable encoding
- sklearn.model_selection.train_test_split — train/test data partitioning
- sklearn.linear_model.LinearRegression — linear regression model
- sklearn.ensemble.RandomForestRegressor — random forest regression model
- sklearn.metrics — r^2 _score, mean_absolute_error, mean_squared_error